AI.203

Ethics in AI and Robotics

COURSE OVERVIEW

Moral and societal dimensions of intelligent machines: bias, safety, accountability, and responsible design.

ETHICS IN AI AND ROBOTICS

1

SESSION

60 MINUTES

ORIENTATION: WHY ETHICS NOW

KEY TOPICS

- Ethics vs. law vs. policy
- AI and robotics impact
- Responsible innovation
- Course framework

TEACHING FOCUS

Treat ethics as a core engineering constraint.

KEY TAKEAWAY Building smart is not enough; it must be right.

2

SESSION

60 MINUTES

BIAS & FAIRNESS

KEY TOPICS

- Sources of bias in data and algorithms
- Fairness metrics
- Mitigation strategies
- Case studies

TEACHING FOCUS

Detect and reduce unfair outcomes.

▲ **KEY TAKEAWAY** Data reflects history—including its injustices.

ETHICS IN AI AND ROBOTICS

3

SESSION

60 MINUTES

SAFETY & RISK

KEY TOPICS

- Functional safety
- Risk assessment
- Fail-safe design
- Autonomous weapons debate

TEACHING FOCUS

Design systems that fail safely.

KEY TAKEAWAY Safety must be engineered in, not bolted on.

ETHICS IN AI AND ROBOTICS

4

SESSION

60 MINUTES

ACCOUNTABILITY & TRANSPARENCY

KEY TOPICS

- Explainable AI (XAI)
- Who is responsible
- Auditability
- Regulatory landscape

TEACHING FOCUS

Make decisions traceable and justifiable.

KEY TAKEAWAY If a robot harms, someone must answer.

5

SESSION

60 MINUTES

PRIVACY & AUTONOMY

KEY TOPICS

- Surveillance and data collection
- Human autonomy and consent
- Workplace automation
- Digital rights

TEACHING FOCUS

Respect persons and their data.

KEY TAKEAWAY Technology should serve human agency, not replace it

ETHICS IN AI AND ROBOTICS

6

SESSION

60 MINUTES

ETHICAL PRACTICE & REVIEW

KEY TOPICS

- Ethical design frameworks
- Course recap and assessment
- Toward robot ethics (AI.402)
- Ethics pledge

TEACHING FOCUS

Adopt a personal code of responsible engineering.

▲ **KEY TAKEAWAY** Engineers are stewards of trust.